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UCE MATHEMATICS 2026
MARKING GUIDE

SECTION A

*Respond to **both** items in this section.*

ITEM 1

A manager of a company organised a sales promotion event in different market centres alongside a fundraising event that took place in the company. The sales promotion event had 27 female and 36 male workers. These were grouped into teams with each team having the same number of males and the same number of females.

The number of market centres visited were equal to the number of teams formed. This promotion increased the company sales by 18% from Shs48,000,000. During the fundraising, the board of directors (BOD) contributed ten million five hundred thousand shillings. The company staff contributed an amount that was $1\frac{1}{2}$ of the amount contributed by the BOD and the sales promotion team contributed 0.04 of the amount obtained after the increased company sales.

The total amount contributed during the fundraising was later donated to specific community schools and charity organisations in the ratio 5:7 respectively.

The BOD has requested the manager to generate information clearly showing the;

I. greatest number of market centres visited during the promotion,

II. number of female and male workers in each team,

III. amount to which the sales increased,

IV. amount donated to schools and charity organisations.

However, the manager has failed to generate the report and needs your help.

Task:
Provide the information needed by the BOD.

RESPONSE

I. Greatest number of market centres visited

The number of market centres equals the number of teams. Every team has the same number of males and the same number of females, so the number of teams must divide both 27 and 36 exactly. The greatest number of teams is therefore the highest common factor (HCF) of 27 and 36.

Expressing each number as a product of prime factors,

$$27 = 3 \times 3 \times 3 = 3^3$$

$$36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$$

The HCF is the product of the common prime factors with the smallest powers,

$$\text{HCF} = 3^2$$

$$\text{HCF} = 9$$

The greatest number of market centres visited = 9.

II. Number of female and male workers in each team

Number of teams = 9.

$$\text{Females in each team} = \frac{27}{9}$$

$$\text{Females in each team} = 3$$

$$\text{Males in each team} = \frac{36}{9}$$

$$\text{Males in each team} = 4$$

Each team has 3 female workers and 4 male workers.

III. Amount to which the sales increased

The sales increased by 18% from Shs48,000,000.

$$\text{Increase} = \frac{18}{100} \times 48{,}000{,}000$$

$$\text{Increase} = 8{,}640{,}000$$

$$\text{New sales} = 48{,}000{,}000 + 8{,}640{,}000$$

$$\text{New sales} = 56{,}640{,}000$$

The sales increased to Shs56,640,000.

IV. Amount donated to schools and charity organisations

Contribution of the BOD:

$$\text{BOD} = 10{,}500{,}000$$

Contribution of the company staff:

$$\text{Staff} = 1\frac{1}{2} \times 10{,}500{,}000$$

$$\text{Staff} = \frac{3}{2} \times 10{,}500{,}000$$

$$\text{Staff} = 15{,}750{,}000$$

Contribution of the sales promotion team:

$$\text{Sales promotion team} = 0.04 \times 56{,}640{,}000$$

$$\text{Sales promotion team} = 2{,}265{,}600$$

Total amount contributed during the fundraising:

$$\text{Total} = 10{,}500{,}000 + 15{,}750{,}000 + 2{,}265{,}600$$

$$\text{Total} = 28{,}515{,}600$$

This total was donated in the ratio 5 : 7, so the number of equal parts is

$$5 + 7 = 12$$

$$\text{Value of one part} = \frac{28{,}515{,}600}{12}$$

$$\text{Value of one part} = 2{,}376{,}300$$

Amount donated to the schools:

$$\text{Schools} = 5 \times 2{,}376{,}300$$

$$\text{Schools} = 11{,}881{,}500$$

Amount donated to the charity organisations:

$$\text{Charity organisations} = 7 \times 2{,}376{,}300$$

$$\text{Charity organisations} = 16{,}634{,}100$$

Check:

$$11{,}881{,}500 + 16{,}634{,}100 = 28{,}515{,}600$$

The total amount donated was Shs28,515,600, of which the schools received Shs11,881,500 and the charity organisations received Shs16,634,100.

Summary of the information for the BOD

Required information	**Result**
Greatest number of market centres visited	9
Workers in each team	3 females and 4 males
Amount to which the sales increased	Shs56,640,000
Total amount donated	Shs28,515,600
Amount donated to schools	Shs11,881,500
Amount donated to charity organisations	Shs16,634,100

ITEM 2

Nalunga has a right-angled triangular garden whose base is 50 m and height is 30 m. She has been informed that she can double her harvest if she expands the garden to a total area of 2400 m² by increasing both the base and the height by the same number of metres without changing the shape of the garden. However, she does not know how to do it.

Nalunga has been buying 40 kg of seeds for planting in her garden from a supplier who has been charging her a total amount of Shs650,000 that included the cost of the seeds as well as a fixed transport cost for delivering the seeds.
With the expanded garden Nalunga will need 60 kg of seeds for planting. The supplier has informed her that she will be charged a total amount of Shs950,000 for both the cost of the seeds and the fixed transport cost for delivering the seeds.

Nalunga's neighbour has a total of Shs500,000 which she intends to spend on both the seeds and their transport for delivery but she is not sure of the number of kilogrammes of seeds the supplier will deliver.

Task:

(a) Determine the;

(i) number of metres by which Nalunga should increase each side of the garden.

(ii) size of the base and the height after the increase.

(b) (i) Form an equation Nalunga and her neighbour would use to determine the amount of money to pay the supplier upon delivery of any number of kilogrammes of seeds.

(ii) Inform Nalunga's neighbour, the number of kilogrammes of seeds the supplier will deliver for the total amount of money she intends to spend.

RESPONSE

(a)(i) Number of metres by which each side should increase

Let the increase in both the base and the height be x metres.

New base $= (50 + x)$ m

New height $= (30 + x)$ m

The new area of the right-angled triangle is 2400 m^2.

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\frac{1}{2}(50 + x)(30 + x) = 2400$$

Multiplying both sides by 2,

$$(50 + x)(30 + x) = 4800$$

Expanding the brackets,

$$1500 + 50x + 30x + x^2 = 4800$$

$$x^2 + 80x + 1500 = 4800$$

Subtracting 4800 from both sides,

$$x^2 + 80x - 3300 = 0$$

Factorising: two numbers whose product is -3300 and whose sum is 80 are 110 and -30.

$$x^2 + 110x - 30x - 3300 = 0$$

$$x(x + 110) - 30(x + 110) = 0$$

$$(x - 30)(x + 110) = 0$$

$$x = 30 \quad \text{or} \quad x = -110$$

The value $x = -110$ is rejected because the garden is being expanded, so the increase must be positive.

Each side should be increased by 30 m.

Note. The original area of the garden is $\frac{1}{2} \times 50 \times 30 = 750$ m^2. The question states the required expanded area directly as 2400 m^2, and this stated value has been used in the working above.

(a)(ii) Size of the base and the height after the increase

$$\text{New base} = 50 + 30$$

$$\text{New base} = 80 \text{ m}$$

$$\text{New height} = 30 + 30$$

$$\text{New height} = 60 \text{ m}$$

Check:

$$\text{Area} = \frac{1}{2} \times 80 \times 60$$

$$\text{Area} = 2400 \text{ m}^2$$

The shape is unchanged because the garden is still a right-angled triangle.

The base is 80 m and the height is 60 m.

(b)(i) Equation for the amount of money to pay the supplier

The total charge consists of the cost of the seeds, which depends on the number of kilogrammes, and a fixed transport cost.

Let n be the number of kilogrammes of seeds, k the cost per kilogramme and F the fixed transport cost. Let C be the total amount paid, in shillings.

$$C = kn + F$$

Using the first purchase, $n = 40$ and $C = 650{,}000$:

$$40k + F = 650{,}000 \qquad (1)$$

Using the expanded garden, $n = 60$ and $C = 950{,}000$:

$$60k + F = 950{,}000 \qquad (2)$$

Subtracting equation (1) from equation (2),

$$60k - 40k = 950{,}000 - 650{,}000$$

$$20k = 300{,}000$$

$$k = \frac{300{,}000}{20}$$

$$k = 15{,}000$$

Substituting $k = 15{,}000$ into equation (1),

$$40(15{,}000) + F = 650{,}000$$

$$600{,}000 + F = 650{,}000$$

$$F = 650{,}000 - 600{,}000$$

$$F = 50{,}000$$

Check using equation (2):

$$60(15{,}000) + 50{,}000 = 900{,}000 + 50{,}000 = 950{,}000$$

The equation is $C = 15{,}000\,n + 50{,}000$, where C is the amount in shillings and n is the number of kilogrammes of seeds.

(b)(ii) Number of kilogrammes the neighbour will receive

The neighbour has Shs500,000, so $C = 500{,}000$.

$$500{,}000 = 15{,}000\,n + 50{,}000$$

Subtracting 50,000 from both sides,

$$450{,}000 = 15{,}000\,n$$

Dividing both sides by 15,000,

$$n = \frac{450{,}000}{15{,}000}$$

$$n = 30$$

Check:

$$15{,}000(30) + 50{,}000 = 450{,}000 + 50{,}000 = 500{,}000$$

The supplier will deliver 30 kg of seeds.

SECTION B

*This section has **two** parts; **I** and **II**.*

Part I

*Respond to **one** item from this part.*

ITEM 3

Hassad and Hajarah are planning to buy scholastic materials from the same shop. Hassad will buy 6 books, 4 pencils and 3 rubbers while Hajarah will buy 8 books, 5 pencils and 4 rubbers.

The price of each book, pencil and rubber is Shs3500, Shs600 and Shs200 respectively. Because prices are expected to increase in the near future, their parent has asked them to arrange the information about the materials as well as their respective prices in rows and columns, so that they request for money to buy **double** each of the materials they need. However, they do not know how to do it.

At the shop there is a back-to-school promotion where after buying scholastic materials, a customer has a chance of winning a gift of either a pen or a ruler by randomly picking two pieces of paper, one at a time, without replacement. If a customer picks two papers with the same item written on them, the customer wins the item.

There are 6 papers with a word **pen** written on them and 5 papers with a word **ruler** written on them.
Hassad and Hajarah wish to determine the probability of winning either a pen or a ruler.

Task:

(a) Organise the information as requested for by the parent.

(b) Determine the amount of money each one should request from their parent.

(c) Inform Hassad and Hajarah their chance of winning either a pen or a ruler.

RESPONSE

(a) Organising the information in rows and columns

The quantities of materials are arranged in a matrix **Q** in which the rows represent the buyers (Hassad, Hajarah) and the columns represent the materials (books, pencils, rubbers).

$$\mathbf{Q} = \begin{pmatrix} 6 & 4 & 3 \\ 8 & 5 & 4 \end{pmatrix}$$

The prices (in shillings) of a book, a pencil and a rubber respectively are arranged in a column matrix **P**.

$$\mathbf{P} = \begin{pmatrix} 3500 \\ 600 \\ 200 \end{pmatrix}$$

To request money to buy **double** each of the materials, the quantity matrix is multiplied by the scalar 2.

$$2\mathbf{Q} = 2\begin{pmatrix} 6 & 4 & 3 \\ 8 & 5 & 4 \end{pmatrix}$$

$$2\mathbf{Q} = \begin{pmatrix} 12 & 8 & 6 \\ 16 & 10 & 8 \end{pmatrix}$$

The information is organised as $2\mathbf{Q} = \begin{pmatrix} 12 & 8 & 6 \\ 16 & 10 & 8 \end{pmatrix}$ **(rows: Hassad, Hajarah; columns: books, pencils, rubbers) and** $\mathbf{P} = \begin{pmatrix} 3500 \\ 600 \\ 200 \end{pmatrix}$**.**

(b) Amount of money each should request

The amount of money required is obtained by multiplying the matrix of doubled quantities by the matrix of prices.

$$\text{Amount} = (2\mathbf{Q})\,\mathbf{P}$$

$$\text{Amount} = \begin{pmatrix} 12 & 8 & 6 \\ 16 & 10 & 8 \end{pmatrix}\begin{pmatrix} 3500 \\ 600 \\ 200 \end{pmatrix}$$

$$\text{Amount} = \begin{pmatrix} 12(3500) + 8(600) + 6(200) \\ 16(3500) + 10(600) + 8(200) \end{pmatrix}$$

$$\text{Amount} = \begin{pmatrix} 42{,}000 + 4{,}800 + 1{,}200 \\ 56{,}000 + 6{,}000 + 1{,}600 \end{pmatrix}$$

$$\text{Amount} = \begin{pmatrix} 48{,}000 \\ 63{,}600 \end{pmatrix}$$

Check for Hassad using the original quantities, then doubling:

$$2[6(3500) + 4(600) + 3(200)] = 2(21{,}000 + 2{,}400 + 600) = 2(24{,}000) = 48{,}000$$

Hassad should request Shs48,000 and Hajarah should request Shs63,600.

(c) Chance of winning either a pen or a ruler

Number of papers with **pen** $= 6$

Number of papers with **ruler** $= 5$

Total number of papers $= 6 + 5 = 11$

A customer wins when the two papers picked, without replacement, have the same word. The customer therefore wins a pen (P, P) or wins a ruler (R, R). These two outcomes are mutually exclusive.

Probability of winning a pen:

$$P(\text{pen}) = P(\text{P first}) \times P(\text{P second})$$

$$P(\text{pen}) = \frac{6}{11} \times \frac{5}{10}$$

$$P(\text{pen}) = \frac{30}{110}$$

$$P(\text{pen}) = \frac{3}{11}$$

Probability of winning a ruler:

$$P(\text{ruler}) = P(\text{R first}) \times P(\text{R second})$$

$$P(\text{ruler}) = \frac{5}{11} \times \frac{4}{10}$$

$$P(\text{ruler}) = \frac{20}{110}$$

$$P(\text{ruler}) = \frac{2}{11}$$

Probability of winning either a pen or a ruler:

$$P(\text{pen or ruler}) = P(\text{pen}) + P(\text{ruler})$$

$$P(\text{pen or ruler}) = \frac{3}{11} + \frac{2}{11}$$

$$P(\text{pen or ruler}) = \frac{5}{11}$$

$$P(\text{pen or ruler}) \approx 0.4545$$

Their chance of winning either a pen or a ruler is $\frac{5}{11}$ (about 0.45).

ITEM 4

A school has organized three seminars in three different terms of the year; the Science seminar, Arts seminar and Mathematics seminar.
Out of the 90 students of senior four, 42 have paid for the Arts seminar while 39 have paid for the Mathematics seminar. 15 students have paid for both the Arts and Mathematics seminars, 12 have paid for both the Science and Mathematics seminars while 21 have paid for both the Science and Arts seminars. The number of students who have paid for the Science seminar only is equal to the number of students who have paid for both the Science and Arts seminars only. Some students have paid for all the seminars. The ratio of the number of students who have paid for all the seminars to the number of students who have not paid for any of the seminars is 3:4.

On the day of the Science seminar, the teacher of English language wants to teach the remaining students only if the probability that a student selected at random from the entire class did not pay for the science seminar, is atleast 0.5.

Task:
Advise the teacher of English language, based on calculations, whether or not to prepare to teach the remaining students.

RESPONSE

Let S, A and M represent the sets of students who paid for the Science, Arts and Mathematics seminars respectively.

$$n(\xi) = 90, \qquad n(A) = 42, \qquad n(M) = 39$$

$$n(A \cap M) = 15, \qquad n(S \cap M) = 12, \qquad n(S \cap A) = 21$$

Let the number of students who paid for all the seminars be x.

$$n(S \cap A \cap M) = x$$

Step 1: Number of students in each region

The numbers given for two seminars include the students who paid for all three, so the numbers who paid for exactly two seminars are:

$$S \text{ and } A \text{ only} = 21 - x$$

$$S \text{ and } M \text{ only} = 12 - x$$

$$A \text{ and } M \text{ only} = 15 - x$$

The number who paid for the Science seminar only equals the number who paid for the Science and Arts seminars only:

$$S \text{ only} = 21 - x$$

Arts only:

$$A \text{ only} = 42 - (21 - x) - (15 - x) - x$$

$$A \text{ only} = 42 - 21 + x - 15 + x - x$$

$$A \text{ only} = 6 + x$$

Mathematics only:

$$M \text{ only} = 39 - (12 - x) - (15 - x) - x$$

$$M \text{ only} = 39 - 12 + x - 15 + x - x$$

$$M \text{ only} = 12 + x$$

The ratio of those who paid for all the seminars to those who paid for none is 3 : 4.

$$\frac{x}{\text{none}} = \frac{3}{4}$$

$$\text{none} = \frac{4x}{3}$$

Step 2: Finding x

The sum of all the regions equals the total number of students.

$$(21 - x) + (21 - x) + (12 - x) + (15 - x) + x + (6 + x) + (12 + x) + \frac{4x}{3} = 90$$

Collecting the numbers and the terms in x separately,

$$(21 + 21 + 12 + 15 + 6 + 12) + (-x - x - x - x + x + x + x) + \frac{4x}{3} = 90$$

$$87 - x + \frac{4x}{3} = 90$$

$$\frac{x}{3} = 90 - 87$$

$$\frac{x}{3} = 3$$

$$x = 9$$

Step 3: Value of each region

Region	**Working**	**Number of students**
Science only	$21 - 9$	12
Science and Arts only	$21 - 9$	12
Science and Mathematics only	$12 - 9$	3
Arts and Mathematics only	$15 - 9$	6

Region	**Working**	**Number of students**
All three seminars	x	9
Arts only	$6 + 9$	15
Mathematics only	$12 + 9$	21
None of the seminars	$\frac{4 \times 9}{3}$	12
Total		**90**

Check against the given information:

$$n(A) = 12 + 6 + 9 + 15 = 42$$

$$n(M) = 3 + 6 + 9 + 21 = 39$$

Step 4: Probability that a student did not pay for the Science seminar

Number who paid for the Science seminar:

$$n(S) = 12 + 12 + 3 + 9$$

$$n(S) = 36$$

Number who did not pay for the Science seminar:

$$n(S') = 90 - 36$$

$$n(S') = 54$$

$$P(\text{did not pay for Science}) = \frac{n(S')}{n(\xi)}$$

$$P(\text{did not pay for Science}) = \frac{54}{90}$$

$$P(\text{did not pay for Science}) = \frac{3}{5}$$

$$P(\text{did not pay for Science}) = 0.6$$

Step 5: Advice

The teacher teaches the remaining students only if the probability is at least 0.5.

$$0.6 > 0.5$$

Advice: the teacher of English language should prepare to teach the remaining students, because the probability that a student selected at random did not pay for the Science seminar is 0.6, which is greater than 0.5. The number of remaining students is 54.

Part II

*Respond to **one** item from this part.*

ITEM 5

John, who lives in town *A*, is a Loans Officer from JK SACCO. He intends to move from home to town *Z* for a meeting and then proceed to town *Y* in the same direction to collect money from a client who took a loan from the SACCO.
Figure **1** shows a sketch map of the location of towns *A*, *Z* and *Y*.

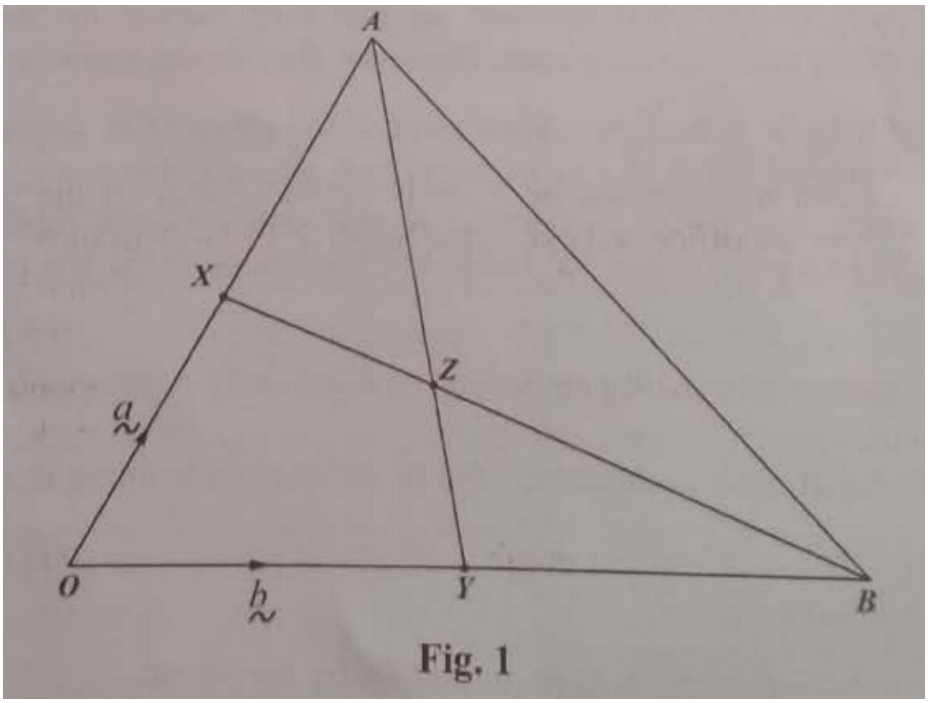


Fig. 1

X and *Y* are places in the middle of the routes *OA* and *OB* respectively. The routes *XB* and *AY* meet at *Z*. *OA* = 2**a**, *OB* = 2**b** and 3*XZ* = *XB*. *AY* is 60 km long.

John needs to use the sketch map and the information given to establish the vector *AZ* in terms of *AY* so as to determine the actual distance of *AZ*.

In addition, John has to be in town *Z* by 8:00 a.m. He is not sure whether he will make it in time if he leaves home at 7:30 a.m, driving at an average speed of 50 km/h. The client at *Y* borrowed Shs3,000,000 at a simple interest rate of 10% per annum for a period of two years. However, the client is doubtful of the total amount of money he should prepare for John to collect.

Task:

(a) Determine the:

(i) relationship between vectors *AZ* and *AY*.

(ii) actual distance of *AZ* in kilometres.

(b) Establish whether John will make it in time for the meeting at *Z*.

(c) Advise the client on the total amount of money he should prepare for John to collect.

RESPONSE

Take O as the origin, with $\overrightarrow{OA}$ and $\overrightarrow{OB}$ given in terms of **a** and **b**.

$$\overrightarrow{OA} = 2\mathbf{a}$$

$$\overrightarrow{OB} = 2\mathbf{b}$$

X is the midpoint of OA and Y is the midpoint of OB, so

$$\overrightarrow{OX} = \frac{1}{2}\overrightarrow{OA} = \mathbf{a}$$

$$\overrightarrow{OY} = \frac{1}{2}\overrightarrow{OB} = \mathbf{b}$$

(a)(i) Relationship between vectors $\overrightarrow{AZ}$ and $\overrightarrow{AY}$

Vector $\overrightarrow{XB}$:

$$\overrightarrow{XB} = \overrightarrow{OB} - \overrightarrow{OX}$$

$$\overrightarrow{XB} = 2\mathbf{b} - \mathbf{a}$$

Since $3XZ = XB$, the point Z divides XB so that $XZ = \frac{1}{3}XB$.

$$\overrightarrow{XZ} = \frac{1}{3}\overrightarrow{XB}$$

$$\overrightarrow{XZ} = \frac{1}{3}(2\mathbf{b} - \mathbf{a})$$

Position vector of Z:

$$\overrightarrow{OZ} = \overrightarrow{OX} + \overrightarrow{XZ}$$

$$\overrightarrow{OZ} = \mathbf{a} + \frac{1}{3}(2\mathbf{b} - \mathbf{a})$$

$$\overrightarrow{OZ} = \mathbf{a} + \frac{2}{3}\mathbf{b} - \frac{1}{3}\mathbf{a}$$

$$\overrightarrow{OZ} = \frac{2}{3}\mathbf{a} + \frac{2}{3}\mathbf{b}$$

Vector $\overrightarrow{AY}$:

$$\overrightarrow{AY} = \overrightarrow{OY} - \overrightarrow{OA}$$

$$\overrightarrow{AY} = \mathbf{b} - 2\mathbf{a}$$

Vector $\overrightarrow{AZ}$:

$$\overrightarrow{AZ} = \overrightarrow{OZ} - \overrightarrow{OA}$$

$$\overrightarrow{AZ} = \frac{2}{3}\mathbf{a} + \frac{2}{3}\mathbf{b} - 2\mathbf{a}$$

$$\overrightarrow{AZ} = -\frac{4}{3}\mathbf{a} + \frac{2}{3}\mathbf{b}$$

Factorising out $\frac{2}{3}$,

$$\overrightarrow{AZ} = \frac{2}{3}(\mathbf{b} - 2\mathbf{a})$$

Since $\overrightarrow{AY} = \mathbf{b} - 2\mathbf{a}$,

$$\overrightarrow{AZ} = \frac{2}{3}\overrightarrow{AY}$$

The relationship is $\overrightarrow{AZ} = \frac{2}{3}\overrightarrow{AY}$. The points A, Z and Y are collinear, and $AZ : ZY = 2 : 1$.

(a)(ii) Actual distance of AZ

$$AZ = \frac{2}{3} \times AY$$

$$AZ = \frac{2}{3} \times 60$$

$$AZ = 40 \text{ km}$$

The actual distance of AZ is 40 km.

(b) Whether John will make it in time for the meeting at Z

John travels from A to Z, a distance of 40 km, at an average speed of 50 km/h.

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\text{Time} = \frac{40}{50}$$

$$\text{Time} = 0.8 \text{ hours}$$

Converting to minutes,

$$\text{Time} = 0.8 \times 60$$

$$\text{Time} = 48 \text{ minutes}$$

John leaves at 7:30 a.m., so his time of arrival at Z is

$$7{:}30 \text{ a.m.} + 48 \text{ minutes} = 8{:}18 \text{ a.m.}$$

$$8{:}18 \text{ a.m.} > 8{:}00 \text{ a.m.}$$

John will not make it in time for the meeting at Z. He will arrive at 8:18 a.m., which is 18 minutes late.

(c) Total amount of money the client should prepare

$$P = 3{,}000{,}000$$

$$R = 10\% \text{ per annum}$$

$$T = 2 \text{ years}$$

Simple interest:

$$I = \frac{P \times R \times T}{100}$$

$$I = \frac{3{,}000{,}000 \times 10 \times 2}{100}$$

$$I = 600{,}000$$

Total amount to be collected:

$$\text{Total} = P + I$$

$$\text{Total} = 3{,}000{,}000 + 600{,}000$$

$$\text{Total} = 3{,}600{,}000$$

The client should prepare Shs3,600,000 for John to collect.

ITEM 6

A monument in form of a pyramid in figure 2 is to be constructed with base PQR, in which PQ = PR = 16 m and QR = 20 m. The vertex, V, is 13 m vertically above P. The outer faces of the monument will be covered with bricks and it is known that 28 bricks cover 1 m².

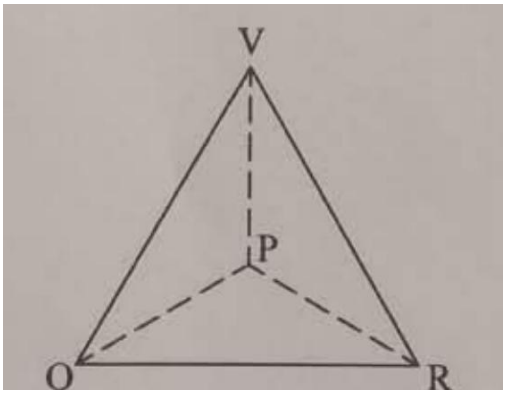


Fig. 2

A manager of a company that has been contracted to construct the monument intends to set off from his office and cycle due North at 15 km/h to move to the site of the monument. At the same time, the company supervisor also intends to set off from her home, 7.5 km due East of the Manager's office and cycle on a bearing of 340° so that she meets the manager before reaching the site of the monument. The supervisor however, does not know how fast she will have to cycle if she is to meet the manager.

The manager also needs to know the number of bricks that will be required for the outer face of the monument.

Task:

(a) Estimate, to the nearest 1000, the number of bricks that will be required.

(b) Recommend, based on calculations, the speed with which the supervisor will travel if she is to meet the manager.

RESPONSE

(a) Number of bricks required

The vertex V is vertically above P, so VP is perpendicular to the base PQR. The base lies on the ground, so the outer faces to be covered are the three sloping faces VPQ, VPR and VQR.

Face VPQ. Since $VP \perp PQ$, triangle VPQ is right-angled at P.

$$\text{Area of } VPQ = \frac{1}{2} \times PQ \times VP$$

$$\text{Area of } VPQ = \frac{1}{2} \times 16 \times 13$$

$$\text{Area of } VPQ = 104 \text{ m}^2$$

Face VPR. Since $VP \perp PR$ and $PR = PQ = 16$ m, triangle VPR is congruent to triangle VPQ.

$$\text{Area of } VPR = \frac{1}{2} \times 16 \times 13$$

$$\text{Area of } VPR = 104 \text{ m}^2$$

Face VQR. Using Pythagoras' theorem in triangle VPQ,

$$VQ^2 = VP^2 + PQ^2$$

$$VQ^2 = 13^2 + 16^2$$

$$VQ^2 = 169 + 256$$

$$VQ^2 = 425$$

In triangle VPR, in the same way, $VR^2 = 425$. Hence $VQ = VR$ and triangle VQR is isosceles. Let N be the midpoint of QR. Then $VN \perp QR$ and

$$QN = \frac{20}{2} = 10 \text{ m}$$

Using Pythagoras' theorem in triangle VNQ,

$$VN^2 = VQ^2 - QN^2$$

$$VN^2 = 425 - 10^2$$

$$VN^2 = 425 - 100$$

$$VN^2 = 325$$

$$VN = \sqrt{325}$$

$$VN = 18.03 \text{ m}$$

$$\text{Area of } VQR = \frac{1}{2} \times QR \times VN$$

$$\text{Area of } VQR = \frac{1}{2} \times 20 \times 18.03$$

$$\text{Area of } VQR = 180.28 \text{ m}^2$$

Total area of the outer faces:

$$\text{Total area} = 104 + 104 + 180.28$$

$$\text{Total area} = 388.28 \text{ m}^2$$

Number of bricks, given that 28 bricks cover 1 m^2:

$$\text{Number of bricks} = 388.28 \times 28$$

$$\text{Number of bricks} = 10{,}871.8$$

Rounding to the nearest 1000,

$$\text{Number of bricks} \approx 11{,}000$$

About 11,000 bricks will be required.

(b) Speed of the supervisor

Let O be the manager's office, H the supervisor's home and M the point where they meet.

The manager cycles due North from O, so M lies due North of O. The home H is 7.5 km due East of O, so angle $HOM = 90°$ and $OH = 7.5$ km.

The supervisor cycles on a bearing of $340°$, which is $20°$ to the west of North ($360° - 340° = 20°$). In triangle OHM, the angle at M between MH and the North-South line MO is therefore $20°$.

$$\angle HOM = 90°$$

$$\angle OMH = 20°$$

$$\angle OHM = 180° - 90° - 20° = 70°$$

Distance cycled by the supervisor, HM:

$$\sin 20° = \frac{OH}{HM}$$

$$HM = \frac{7.5}{\sin 20°}$$

$$HM = \frac{7.5}{0.3420}$$

$$HM = 21.93 \text{ km}$$

Distance cycled by the manager, OM:

$$\tan 20° = \frac{OH}{OM}$$

$$OM = \frac{7.5}{\tan 20°}$$

$$OM = \frac{7.5}{0.3640}$$

$$OM = 20.61 \text{ km}$$

Time taken by the manager to reach M:

$$\text{Time} = \frac{OM}{\text{speed}}$$

$$\text{Time} = \frac{20.61}{15}$$

$$\text{Time} = 1.374 \text{ hours}$$

They set off at the same time and meet at M, so the supervisor also takes 1.374 hours.

$$\text{Speed of the supervisor} = \frac{HM}{\text{Time}}$$

$$\text{Speed of the supervisor} = \frac{21.93}{1.374}$$

$$\text{Speed of the supervisor} = 15.96 \text{ km/h}$$

Check using the ratio of the distances, since the times are equal:

$$\frac{v}{15} = \frac{HM}{OM}$$

$$v = 15 \times \frac{1}{\cos 20°}$$

$$v = \frac{15}{0.9397}$$

$$v = 15.96 \text{ km/h}$$

The supervisor should cycle at about 16 km/h (15.96 km/h) in order to meet the manager.